Survival-supervised deconvolution identifies a replicable tumor–stroma prognostic architecture in pancreatic cancer

Amber M. Young [a], Alisa Yurovsky [b], Xianlu Laura Peng [c,d], Jen Jen Yeh [c,d,e], Didong Li [a], Naim U. Rashid [a,d,*]

[a]Department of Biostatistics, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA

[b]Department of Biomedical Informatics, Stony Brook University, Stony Brook, NY 11794, USA

[c]Department of Pharmacology, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA

[d]Lineberger Comprehensive Cancer Center, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA

[e]Department of Surgery, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA

[*]Corresponding author: nur2@email.unc.edu

Abstract

Pancreatic ductal adenocarcinoma (PDAC) outcome reflects both malignant and stromal state, yet clinically relevant transcriptomic programs are typically discovered without outcome information and tested against survival only afterward. We developed DeSurv, a survival-supervised matrix factorization that incorporates clinical outcome into gene-program estimation while holding the learned gene weights fixed for evaluation in new cohorts. In pooled TCGA and CPTAC PDAC cohorts, DeSurv recovered a three-program tumor–stroma architecture in which a dominant survival-aligned direction linked classical tumor identity with restCAF-like stroma, alongside separate proCAF-like stromal and basal-like tumor programs. The frozen representation transferred across five external validation datasets spanning four independent patient cohorts, with the combined DeSurv risk score remaining prognostic (HR per SD, 1.42; 95% CI, 1.26–1.59), including after PurIST and DeCAF adjustment, and D1 retaining prognostic relevance in treated metastatic PDAC. Unsupervised factorization matched on the number of programs and the tuning budget did not recover the survival-aligned direction, indicating that it arose from survival supervision rather than model size or computational effort.

Pancreatic ductal adenocarcinoma (PDAC) is one of the most lethal solid tumors, with few durable responses to existing therapies[1]. Bulk transcriptomic studies have identified clinically relevant malignant and stromal states in PDAC, including basal-like and classical tumor programs[2–4] and proCAF and restCAF stromal programs, defined respectively as tumor-promoting and tumor-restraining[5]. Single-cell and spatial profiling have further shown that this heterogeneity extends across malignant, stromal, immune, and other microenvironmental compartments within individual tumors[6–8]. Whether these tumor and stromal programs together provide prognostic information beyond existing subtype classifications remains unclear.

Although single-cell and spatial profiling resolve cellular states in detail, large clinically annotated PDAC cohorts still rely primarily on bulk transcriptomic profiling[9,10]. Bulk profiles provide the scale needed for survival analysis but mix signals from malignant and microenvironmental cells[11]; nonnegative matrix factorization (NMF) and related deconvolution methods can separate these composite profiles into interpretable gene programs, weighted sets of co-varying genes that together form a representation of the tumor transcriptome[3,12–14]. In PDAC, such methods have yielded molecular subtypes[2,4,15], virtual microdissection of basal-like and classical malignant programs[3], and compartment-specific tumor–stroma deconvolution[5,14]. The practical challenge is therefore not simply to resolve interpretable gene programs, but to identify during discovery which programs are clinically relevant and to test those same programs unchanged across independent studies.

Such deconvolution approaches typically discover gene programs without using clinical outcomes and assess their prognostic relevance only afterward[2,15]. Standard NMF learns gene programs by minimizing reconstruction error, the mismatch between the measured expression matrix and the approximation rebuilt from the programs, favoring patterns that capture prominent expression variation whether or not that variation is related to survival[9,16]. This distinction is particularly important in PDAC, where bulk tumors contain substantial stromal, exocrine, and other non-malignant signals that can dominate transcriptomic variation and obscure less prominent but prognostically relevant programs[4,5,17,18]. Reconstruction alone also does not uniquely determine the recovered gene programs without additional constraints[19–21], which may contribute to cross-study variability among unsupervised PDAC subtype taxonomies[22]. Consequently, determining which unsupervised programs are both clinically relevant and transportable across studies has required extensive multi-study refinement. In PDAC, the basal-like/classical dichotomy was established and progressively validated over roughly a decade through iterative subtyping, virtual microdissection, independent bulk cohorts, single-cell analyses, and prospective clinical profiling[2–4,8,10]; comparable work is now establishing the clinical relevance of stromal CAF subtypes[5].

Conversely, survival-supervised models such as penalized Cox regression and supervised principal components[9,23] can identify outcome-associated expression patterns, but they often operate on thousands of highly correlated genes, making individual-gene effects difficult to estimate and interpret. Such models may predict survival without organizing the prognostic signal into biologically interpretable tumor and microenvironmental programs, which is the form in which PDAC subtypes, and the single-sample classifiers built from them, have been defined[4,5]. A risk score alone does not supply that substrate. Together, these limitations motivate an approach that incorporates survival information as gene programs are learned, while preserving their biological interpretability.

Here, we present DeSurv, a survival-supervised deconvolution framework that integrates NMF with Cox regression so that survival helps shape gene programs as they are learned. By incorporating survival during factorization, DeSurv is designed to recover survival-aligned biological structure that may be obscured when other sources of expression variation dominate. Once learned, the gene weights defining each program are held fixed, or frozen, allowing the same programs to be evaluated directly in independent cohorts without refitting the factorization. DeSurv therefore changes clinical relevance from a post hoc criterion used to select among unsupervised factors into a criterion that shapes the factors themselves, while making cross-study transportability a direct test of the same frozen gene programs. We evaluate DeSurv in PDAC across five external validation datasets and use simulations in which the true underlying (latent) programs are known to characterize when survival supervision improves recovery of prognostic programs.

In PDAC, DeSurv identifies a three-program tumor–stroma architecture that reflects joint cross-patient variation rather than spatial or cellular organization within individual tumors. We compare it with NMF at higher rank, where the rank is the number of programs the factorization extracts, and with conventional survival-supervised models to isolate what survival supervision contributes to the representation itself.

Results

DeSurv incorporates survival information into gene-program estimation

DeSurv learns gene programs by balancing reconstruction of the tumor transcriptome with association to patient survival. Unlike the conventional sequence of unsupervised factorization followed by post hoc clinical screening, survival directly shapes the gene programs during discovery. After training, the gene weights are fixed, allowing the same programs to be scored and tested in new tumors without refitting the factorization (Fig. 1A; Methods). We trained DeSurv in pooled TCGA and CPTAC PDAC cohorts[24,25] ($n = 273$, 139 events). Settings were chosen by Bayesian optimization[26,27], a sequential search algorithm that proposes each new setting from the results of previous ones to optimize a pre-specified objective (Methods). Cross-validated concordance (the C-index, the probability that the model ranks a pair of patients in the correct survival order) was high across a broad region of intermediate supervision and moderate rank, declining toward the unsupervised $\alpha = 0$ boundary (Supplementary Fig. S1). Concordance changed little across the ranks examined, so we applied the one-standard-error rule, which takes the most parsimonious model within one standard error of the best, giving a three-program model ($k = 3$, $\alpha = 0.334$; peak C-index 0.655; Fig. 1B and Supplementary Fig. S4). By contrast, standard unsupervised NMF rank criteria (reconstruction residuals, cophenetic correlation and silhouette width[13,28]) gave conflicting choices in the same data (Supplementary Fig. S5). All model and hyperparameter selection was completed within the training cohorts before the programs were applied to external validation datasets.

Survival supervision resolves a distinct tumor–stroma program structure

To determine how survival supervision altered the biological representation, we compared DeSurv with standard NMF at the same rank ($k = 3$), thereby holding model complexity constant. DeSurv resolved three distinct programs (D1–D3): D1, enriched for classical tumor and restCAF-like stromal signatures; D2, enriched for proCAF-like stroma; and D3, enriched for basal-like tumor programs (Fig. 2a). Together, these programs define a tumor–stroma prognostic architecture spanning classical/restCAF-like and basal-like/proCAF-like states across patients. The classical identity of D1 was independently supported in the COMPASS cohort, where D1 scores correlated with GATA6 RNA in situ hybridization (Spearman $\rho = 0.68$, $P < 0.001$, $n = 33$; Fig. 2e).

Standard NMF produced a different representation at the same rank (Fig. 2b). Its malignant factor (N1) was enriched for both classical and basal-like tumor programs, its microenvironmental factor (N3) for multiple stromal and immune programs[29,30], and a third factor (N2) for exocrine-associated variation. DeSurv instead separated basal-like tumor (D3) and proCAF-like stroma (D2) while recovering the distinct classical/restCAF-like survival-aligned D1 program. Both corresponded comparably to the activated-stroma and proCAF-like reference programs, the stromal distinctions bulk deconvolution conventionally resolves, but differed at the restraining end: the restCAF and iCAF reference programs correlated with a DeSurv program, and with no standard NMF factor at the same rank (Fig. 2a,b).

This difference reflected a selective reorganization rather than wholesale loss of expression structure. D1 accounted for the smallest share of the full model's reconstruction gain, 18.8% (Shapley shares, which sum to 100% across the three programs[31]), yet carried essentially all of the survival signal (Fig. 2c): omitting it lowered the Cox partial log-likelihood, the fit criterion of the Cox model, by 66.4, against 1.0 for D3 and 0.02 for D2, even though D3 accounted for 56% of it. Because D1 was estimated using these outcomes, its increment describes where the model placed the survival signal rather than testing whether that signal is real. The unsupervised factors are not subject to the same concern: they were fit without survival, and the largest reconstructor among them, at 42% of the reconstruction gain, lowered the partial log-likelihood by only 1.16. The programs that best reconstruct the transcriptome need not be those most relevant to survival[18].

Correlating each NMF factor's gene loadings with each DeSurv factor's, across all analysed genes, showed that the principal NMF tumor and microenvironmental factors were largely retained as D3 (Spearman $\rho = 0.84$) and D2 ($\rho = 0.87$), whereas D1 correlated only weakly with every matched-rank NMF factor ($\rho \leq 0.36$; Fig. 2d). Consistently, D1 was poorly reconstructable from the full set of NMF factors ($R^2 = 0.09$; Supplementary Table S2), while reconstructing the analysis matrix somewhat less completely than unsupervised NMF (R^2 0.75 versus 0.81, a difference of 6.0 percentage points; Supplementary Table S3). Survival supervision therefore preserved most of the variance-dominant tumor and stromal structure while resolving a distinct survival-aligned program, at a measurable cost in reconstruction. Tuning effort does not appear to explain the difference. We repeated the comparison with an unsupervised model at the same rank, given DeSurv's optimizer, its tuning budget and its consensus initialization (starting values pooled over many restarts). It recovered D2 ($\rho = 0.86$) and D3 ($\rho = 0.75$) as before, but recovered D1 no better than the standard implementation did ($\rho \leq 0.24$).

Frozen survival-aligned programs retain prognostic relevance across independent untreated and treated PDAC cohorts

We next applied the frozen programs to five external validation datasets spanning four independent patient cohorts (Dijk, Moffitt, PACA-AU [array and RNA-seq], and Puleo; non-metastatic, treatment-naive; Supplementary Table S1). Each tumor was scored by projecting its expression onto the trained programs, restricted to their top-ranked genes ($Z = \widetilde{W}^\top X_{\text{new}}$; Methods). Combined analyses represented each patient once, giving 570 unique patients (388 deaths; Methods). The survival-aligned D1 program was protective in every validation dataset and was the only individual program associated with survival in the combined, cohort-stratified analysis (HR per SD 0.75 (0.68–0.83), P <0.001). D3 pointed toward higher risk in four of the five datasets but was not significant in any of them, nor combined (1.09 (0.98–1.21), $P = 0.10$), and the proCAF-like D2 program was near-null (1.02 (0.92–1.12), $P = 0.76$) and more variable (Fig. 3a). The frozen multivariable DeSurv linear predictor, the weighted sum of the three program scores (oriented so that higher values indicate higher risk), was associated with overall survival in a cohort-stratified analysis (pooled HR per SD = 1.42, 95% CI 1.26–1.59, $P = 5.4 \times 10^{-9}$; Table 1). The estimate did not depend on any single dataset (Fig. 3a). At the same complexity, unsupervised predictors were not associated with survival, whether tuned by DeSurv's own budget or not (Table 1).

We then asked whether comparable prognostic performance could be achieved without survival-supervised deconvolution. The two methods differ most in how their performance depends on rank (Supplementary Fig. S4). DeSurv reaches essentially its full cross-validated concordance at the smallest ranks examined and is flat thereafter, so added factors buy it nothing. Unsupervised NMF starts well below DeSurv at low rank and climbs steadily as factors are added, closing most but not all of the gap by the largest rank examined; across this range its cross-validated concordance does not reach the value DeSurv attains with three factors. The two curves meet at a single rank, and there because the DeSurv curve dips rather than because NMF gains. Unsupervised factorization therefore requires many more factors to approach the same prognostic information, and at the rank where it does best in external validation those factors redundantly overlap the same established signatures rather than resolving additional biology (Supplementary Fig. S6). Survival supervision is thus not required to recover prognostic signal, but it concentrates that signal into a decomposition less than half the size (Table 1; Supplementary Table S5).

Table 1: Pooled external validation hazard ratios (per SD of the linear predictor) for DeSurv at $k = 3$ and standard NMF at BO-selected $k = 7$ ($\alpha = 0$), across five validation datasets spanning four independent patient cohorts (combined $n = 570$ unique patients, 388 deaths). Adjusted models include the PurIST tumor-intrinsic and DeCAF stromal molecular classifiers as covariates, with stratification by dataset, and are restricted to the $n = 570$ patients (388 deaths) with both classifier calls available. Both methods retain significant prognostic value after adjustment. Both unsupervised comparators shown here are the unsupervised limit of the same optimizer, run at $\alpha = 0$ with the identical consensus initialization and Bayesian-optimization budget given to the supervised model: one at its own BO-selected rank ($k = 7$) and one held at DeSurv's rank ($k = 3$), the latter serving as the tuning-budget control for the representational comparison in Fig. 2. The matched-rank comparator used for the factor-structure panels of Fig. 2 is Lee multiplicative NMF, the field-standard implementation (Supplementary Table S5).

Method	Unadjusted HR per SD (95% CI), P	Adjusted HR per SD (95% CI), P
DeSurv ($k = 3$)	1.42 (1.26–1.59), < 0.001	1.28 (1.11–1.47), < 0.001
Unsupervised, matched rank ($k = 3$, $\alpha = 0$)	1.04 (0.94–1.16), 0.428	0.98 (0.88–1.09), 0.705
Unsupervised NMF ($k = 7$, $\alpha = 0$)	1.44 (1.30–1.59), < 0.001	1.40 (1.22–1.62), < 0.001

The programs above were learned in treatment-naive, non-metastatic disease. Projected without retraining into independently treated PDAC cohorts, the survival-aligned D1 program remained associated with overall survival in treated metastatic disease under an arm-stratified analysis, and the proCAF-like D2 program increased after treatment (Fig. 4; Supplementary Information, Translational transportability).

The survival-aligned structure is not method-specific and is complementary to established PDAC classifiers

We next asked whether the dominant D1 direction simply recapitulated established PDAC subtype classifications. Across the pooled validation cohorts ($n = 570$, 388 events), binary PurIST and DeCAF classifications together explained 34% of the variance in the continuous D1 score (ordinary R^2 from a linear model with cohort fixed effects, fitted to D1 scores already standardized within cohort), indicating substantial alignment with known tumor-intrinsic and stromal subtypes but incomplete overlap. D1 remained associated with survival after simultaneous adjustment for both classifiers (HR per SD = 0.82, 95% CI 0.73–0.93, $P = 0.0014$); adding D1 to a stratified Cox model containing PurIST and DeCAF improved model fit (partial likelihood-ratio $\chi^2 = 10.1$ on 1 df, $P = 0.0015$) and concordance from 0.592 to 0.622. The DeSurv linear predictor also remained prognostic after adjustment for standard clinical covariates. Because D1 spans tumor and stromal programs, we asked whether it tracks tumor purity. Where purity was measured on a full cohort the association was essentially unchanged (PACA-AU array, 1.44 unadjusted versus 1.45 adjusted;

RNA-seq, 1.66 versus 1.65), arguing against a purity artifact (Supplementary Table S4). Thus, the survival-aligned representation overlaps established tumor and stromal classifications without being reducible to them.

The dominant D1 direction was also not specific to the DeSurv optimizer. Three independent survival-supervised approaches, in simple single-component form (supervised principal components, Cox partial least squares and sparse Cox), recovered patient scores strongly correlated with D1 in the training cohort ($|r| = 0.81$, 0.83 and 0.85, respectively), whereas the leading unsupervised principal component was only weakly correlated with D1 ($|r| = 0.10$) and instead tracked higher-variance D2/D3 structure (Supplementary Information, The D1 axis is recovered by survival supervision, not by DeSurv specifically). Thus, D1 reflects a survival-associated expression direction shared across supervised approaches rather than an idiosyncrasy of DeSurv.

Having established that D1 is not specific to DeSurv, we asked what DeSurv adds beyond recovering it. Independently tuned supervised principal components and penalized Cox achieved pooled external concordance comparable to DeSurv (C-index 0.60, 0.60 and 0.61, respectively) and captured the dominant D1-associated prognostic direction. Their selected genes, however, were consistently enriched for basal-like tumor programs but not proCAF, restCAF, immune or exocrine programs, and their risk scores were weakly correlated with the separately resolved proCAF-like D2 program (penalized Cox, $|r| = 0.06$; supervised PCA, $|r| = 0.14$; Fig. 3b). Although these scores retained modest stromal signal, they did not resolve it as a distinct stromal program (Supplementary Information). DeSurv instead places that direction within a fixed set of tumor and stromal programs that can be measured and interpreted separately, and applied unchanged to new cohorts.

DeSurv recovers prognostic gene programs when variance and prognosis diverge

We used simulations with known latent structure to characterize when survival supervision alters program recovery. In the primary scenario, the gene program driving survival contributed less transcriptomic variation than outcome-neutral programs ($p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$; Materials and Methods, Simulation Studies), so that the strongest sources of expression variation were not the most prognostically relevant. DeSurv and standard NMF ($\alpha = 0$) were tuned using the same cross-validation procedure, isolating the effect of survival supervision during factorization.

Under this setting, DeSurv more accurately recovered the genes defining the true prognostic program, whereas standard NMF largely followed variance-dominant structure (Fig. 5a–b). This distinction is important because the genes assigned to a recovered program, not only its ability to rank patients by risk, determine its subsequent biological interpretation and its potential use in candidate signatures or biomarker development. Across 100 replicates, median test-set C-index was 0.837 for DeSurv versus 0.724 for standard NMF ($\Delta = 0.113$, paired Wilcoxon $P < 0.001$), and DeSurv recovered the true factorization rank ($k = 3$) more frequently (Fig. 5c).

The advantage depended on the relationship between transcriptomic variance and prognosis. It attenuated when prognostic and variance-dominant structure partially overlapped and disappeared under the null, where neither method identified prognostic signal (Supplementary Fig. S3). In the mixed setting, the advantage remained strongest for recovery of the factor-specific marker genes rather than shared background survival genes, supporting a specific effect of supervision on identifying the genes that define the prognostic program.

Discussion

In pancreatic cancer, survival-supervised factorization identified a dominant survival-aligned program, D1, linking classical tumor identity to restCAF-associated stromal biology, alongside separately resolved proCAF-like stromal and basal-like tumor programs. D1 was recovered de novo without subtype labels, was supported orthogonally by GATA6 RNA in situ hybridization, and remained prognostic after adjustment for the established PurIST and DeCAF classifications, so it is not reducible to them. The trained programs then transferred without refitting across five external validation datasets spanning four independent patient cohorts. Its contribution was to place this replicable prognostic direction within a biologically decomposed tumor–stroma architecture, rather than returning a variance-driven factorization or a single supervised risk score. The advance is in how clinically relevant programs are discovered, not in how well they predict.

The distinction was not one of model size, tuning effort or predictive accuracy. An unsupervised model given DeSurv's rank, optimizer, tuning budget and consensus initialization still failed to recover D1. Higher-rank NMF and conventional supervised predictors each recovered the prognostic information while representing its biology differently, the former distributing established signatures across overlapping factors, the latter resolving a tumor-intrinsic axis without the stromal programs. In simulations, where the latent programs are known, DeSurv more accurately recovered the genes defining the true prognostic program.

This addresses a practical limitation of the conventional discovery sequence. When programs are learned without reference to outcome, their clinical relevance and correspondence across datasets must be established retrospectively through factor selection, biological annotation and cross-study matching. DeSurv does not eliminate the need for external validation, but makes that validation a direct test of programs already shaped by clinical relevance and held fixed across studies.

This distinction is consistent with the structure of the DeSurv objective. Survival supervision acts directly on the gene-program matrix W that holds those weights, while the reconstruction objective requires those same programs to continue representing the observed transcriptome. Once learned, the weights are fixed, and a new tumor is scored by projecting its expression profile onto the top-ranked genes of each program, so the same programs can be evaluated across cohorts without re-estimating the factorization. Empirically, supervision largely retained the major variance-driven tumor and stromal structure represented by D3 and D2 while introducing a D1 program poorly reconstructable from the matched-rank NMF factors, at a measurable cost in whole-matrix reconstruction. These findings are consistent with survival supervision selectively reorganizing rather than replacing the variance-driven representation. Prior survival-representation approaches, including survival-supervised matrix factorization[32,33] and deep latent-variable models[34,35], have also integrated outcome information with learned representations, but generally do not preserve the same directly transferable gene-program decomposition.

Biologically, D1 represents the dominant survival-aligned program within the broader tumor–stroma architecture rather than the architecture itself. It links classical tumor identity with restCAF-like stromal biology and was recovered de novo without predefined subtype labels. Its classical identity was supported orthogonally by GATA6 RNA in situ hybridization, and its survival association persisted after adjustment for PurIST and DeCAF. The additional D2 and D3 programs provide separately measurable stromal and tumor context around this direction, rather than necessarily representing independent components of the fitted risk score. This architecture is consistent with growing evidence that PDAC outcome reflects interactions between malignant lineage state and heterogeneous fibroblast and stromal programs[36–38].

Rank selection represents an additional source of uncertainty in unsupervised decomposition. In the PDAC training data, commonly used NMF criteria gave conflicting rank choices, and increasing rank progressively redistributed established biological signatures across additional factors. Together with the non-uniqueness of reconstruction-based factorization, this rank dependence may contribute to variability among unsupervised molecular taxonomies across studies[2,3,15]. Survival supervision does not remove that non-uniqueness, and single DeSurv runs from different random starts recover appreciably different gene sets. What makes the reported decomposition a stable object is the consensus step, and independently seeded replicates of the full procedure recover it closely (Supplementary Information, Stability of the recovered gene programs). Survival-supervised selection therefore does not eliminate factorization uncertainty, but it provides an outcome-relevant criterion for choosing among representations when the scientific objective is specifically to identify clinically relevant programs.

Our simulations clarify when this approach is most useful. When the genes defining the prognostic program contributed relatively little to overall transcriptomic variation, survival supervision improved their recovery; this advantage diminished as prognostic and variance-dominant structure overlapped and disappeared when no survival signal was present. Correct program recovery matters beyond discrimination because the genes assigned to a program determine downstream pathway interpretation, candidate signatures and potential biomarker development.

DeSurv is also distinct from reference-based methods such as CIBERSORTx[39] and BayesPrism[40], which estimate cellular composition using external cell-state references. Instead, DeSurv learns outcome-aligned latent gene programs directly from bulk expression and could in principle be combined with reference-based cell-fraction estimates to refine their cellular attribution.

Several limitations define the scope of these findings. DeSurv currently uses a proportional-hazards survival objective and was evaluated here only in PDAC. Extensions to non-proportional or nonlinear survival relationships are therefore natural methodological directions. More importantly, the simulations establish conditions under which survival supervision improves recovery of prognostic programs, but whether the same representational advantage generalizes across cancer types remains unknown. Testing it will require multi-cancer benchmarking against unsupervised decompositions and against strong penalized-regression baselines, which remain difficult to outperform consistently for out-of-sample prognosis[41].

The treated-cohort analyses reported above are exploratory. Within the O'Kane/COMPASS cohort the individual treatment arms were too small to show a consistent treatment-specific survival pattern, whereas the prognostic association held in the better-powered arm-stratified analysis. Whether these programs predict differential treatment benefit therefore remains unresolved and will require adequately powered, prespecified treatment-by-program interaction analyses.

The tumor–stroma architecture described here is a bulk-tissue phenomenon. It reflects cross-patient covariation of tumor- and stroma-associated transcriptional programs, not cellular co-expression or spatial adjacency within individual tumors. The biological identity of the survival-aligned D1 program is nevertheless supported by its persistence after PurIST, DeCAF and tumor-purity adjustment, its recovery by independent survival-supervised approaches, and its correlation with GATA6 RNA in situ hybridization. Mapping these bulk programs onto specific cellular states is not a matter of projecting the same gene weights into single-cell or spatial data. Most of the genes defining each program go undetected in any individual cell, so per-cell projection scores partly reflect detection depth rather than program activity, and multicellular spatial spots mix the compartments the programs are meant to separate. Establishing the cellular correlates of these programs will therefore require clinically annotated single-cell and spatial cohorts together with scoring methods designed for sparse measurement.

More broadly, unsupervised transcriptomic discovery is optimized to recover what varies most, whereas clinically supervised prediction is optimized to recover what predicts outcome; these need not be the same biological structure. DeSurv addresses this gap by incorporating survival into program discovery while retaining an explicitly decomposed gene-program representation. In PDAC, this revealed a replicable tumor–stroma prognostic architecture that could be transferred across cohorts, biologically validated and interrogated beyond a risk score alone. By making clinical relevance part of discovery and cross-study transportability a test of the same frozen gene programs, it complements unsupervised deconvolution and conventional supervised prediction without depending on superior predictive accuracy.

Methods

Problem formulation and notation

Let $X \in \mathbb{R}_{\geq 0}^{p\times n}$ denote the nonnegative gene expression matrix. DeSurv approximates $X \approx WH$, where $W \in \mathbb{R}_{\geq 0}^{p\times k}$ contains nonnegative gene programs and $H \in \mathbb{R}_{\geq 0}^{k\times n}$ contains sample loadings. Additionally, let $y \in \mathbb{R}_{\geq 0}^{n}$ denote patient survival times, $\delta \in \{0,1\}^n$ the event indicators ($\delta_i = 1$ for an observed event, $\delta_i = 0$ for censoring), and $\beta \in \mathbb{R}^k$ the Cox regression coefficients. Survival outcomes are modeled through a Cox proportional hazards model with factor scores $Z = W^\top X \in \mathbb{R}^{k\times n}$: for sample i with projected score $z_i = W^\top x_i$, the hazard is $h(t \mid x_i) = h_0(t)\exp(z_i^\top \beta)$, where h_0 is the baseline hazard. Defining factor scores by projection onto the learned gene programs ($Z = W^\top X$) rather than by the inferred loadings H lets the same gene weights score new cohorts without refitting, the property we exploit for external validation.

The DeSurv Model

DeSurv integrates NMF with penalized Cox regression to identify gene programs associated with survival outcomes. The method operates in a survival-supervised framework: the reconstruction loss penalizes deviation from the observed expression data, while the survival term directs gene programs toward outcome-relevant structure. The parameter $\alpha \in [0, 1)$ controls the relative contribution of each objective.

The joint objective is

$$\mathcal{L}(W, H, \beta) = (1-\alpha)\,\mathcal{L}_{\mathrm{NMF}}(W, H) - \alpha\,\mathcal{L}_{\mathrm{Cox}}(W, \beta), \tag{1}$$

where $\mathcal{L}_{\mathrm{NMF}}(W, H)$ is the NMF reconstruction error (including an ℓ_2 penalty on H) and $\mathcal{L}_{\mathrm{Cox}}(W, \beta)$ is the elastic-net penalized partial log-likelihood. A key design choice is where survival supervision enters the factorization. Because factor scores are defined as $Z = W^\top X$, the Cox partial likelihood $\mathcal{L}_{\mathrm{Cox}}$ is a direct function of W and β, and the survival gradient acts explicitly on the gene program matrix W. The sample-level loadings H enter only through the reconstruction term and receive no survival gradient, preserving their interpretation as mixture coefficients[12,42]. This construction makes clinical outcome part of program estimation rather than a post hoc filter applied to an unsupervised decomposition. When $\alpha = 0$, DeSurv reduces to standard unsupervised NMF, and β

is fit post hoc by penalized Coxnet on the projected scores $Z = W^\top X$, used only for scoring and cross-validation without affecting the factorization. The Cox component also supports optional adjustment for sample-level covariates (for example, stage or grade); the analyses reported here use factor scores alone.

Optimization alternates updates for H, W, and β (Supplementary Information, Section 2), with backtracking used for proposed W updates. The reconstruction term is minimized using multiplicative updates that preserve nonnegativity; the supervision term enters W's gradient as a Cox partial-likelihood derivative; and β is updated by Coxnet coordinate descent on the current factor scores $Z = W^\top X$ with elastic-net penalty parameters (λ, ξ). Because the implemented algorithm combines normalization, gradient balancing and a Coxnet approximation, we assessed optimization behavior empirically rather than relying on a formal convergence guarantee: across independent random initializations of the reported PDAC model, all 100 runs terminated on the stopping criterion and the fitted objective decreased monotonically to numerically stable solutions (Supplementary Information, Empirical optimization stability). Because objective convergence does not imply that the same programs are recovered each time, we separately assessed program-level stability. Individual restarts are dispersed, whereas the reported consensus procedure recovers that decomposition closely across independently seeded replicates (Supplementary Information, Stability of the recovered gene programs). Closed-form update equations are given in Supplementary Information, Sections 1–2, and Bayesian-optimization control parameters in the Bayesian Optimization section.

Hyperparameter selection and cross-validation

Hyperparameters $(k, \alpha, \lambda, \xi, n_{\text{top}})$ were selected by maximizing the cross-validated Harrell C-index using Bayesian optimization (BO)[26,27], which finds near-optimal configurations using far fewer objective evaluations than grid or random search over mixed integer–continuous hyperparameter spaces, with final rank k chosen by the one-standard-error rule (the smallest k whose predicted performance lies within one standard error of the maximum). All model selection was performed entirely within the training data using cross-validation; no validation cohort data were used during any stage of tuning, and unbiased generalization was assessed only on the external validation cohorts. Each fold was trained using multiple random initializations, and fold-level performance was defined as the average C-index across initializations. For stability, we used a consensus-based initialization for the final model, aggregating multiple DeSurv runs into a gene-gene co-occurrence matrix and constructing an initialization W_0 from the resulting clusters (Supplementary Information, Consensus initialization for the final model). Before projection, the gene support was restricted to the union across factors of each program's BO-selected top genes, ranked by factor specificity, giving the truncated matrix $\widetilde{W}$ that was held fixed for all validation cohorts. Two scoring conventions follow from this support and are used for different quantities. Factor-level scores retain all k columns, $Z = \widetilde{W}^\top X_{\text{new}}$, and are used for the per-factor and pooled survival analyses. Where a single risk score is required, as in cross-validated tuning, the columns are first collapsed to $\theta = \widetilde{W}\beta$ and θ is scaled to unit ℓ_2 norm before projection. The two differ only by a positive scalar and so give identical concordance, but we state both because the normalization applies to the collapsed score rather than to individual columns of W. In PDAC, BO selected $n_{\text{top}} = 270$ top genes per factor. The ridge penalty on the loadings (H) was held fixed at 0 in all analyses and was not tuned; factor scaling was instead controlled directly by column-normalization of W during optimization.

Because the goal is prognostic factorization, DeSurv selects k using the criterion that directly measures prognostic performance (cross-validated C-index) rather than the unsupervised diagnostics commonly used for NMF rank selection[13,28]. In the PDAC training data, standard NMF diagnostics (reconstruction residuals, cophenetic correlation, mean silhouette width) yielded inconsistent guidance across candidate ranks, a known limitation of unsupervised rank-selection criteria that can disagree on the same dataset (Supplementary Fig. S5). Cross-validated concordance varied little across candidate ranks, with no clear optimum anywhere between $k = 2$ and $k = 12$ and a total spread across that range of about two standard errors (Supplementary Fig. S4). The one-standard-error rule applied to joint BO over rank and supervision strength accordingly selected the parsimonious $k = 3$ at $\alpha = 0.334$. The three-program solution should therefore be read as a parsimonious choice within a flat performance region, with its biological support coming from reference-program correspondence and the GATA6 comparison rather than from a sharp statistical optimum.

External validation cohorts were scored by projecting new datasets onto the learned programs via $Z = \widetilde{W}^\top X_{\text{new}}$, and survival associations were evaluated using Harrell's C-index, hazard ratios and log-rank statistics. Harrell's C-index is rank-based and assesses discrimination only, not calibration or time-dependent prediction error, so we additionally report a proper score as a sensitivity check: the integrated Brier score and its index of prediction accuracy relative to a Kaplan–Meier reference (Supplementary Information, Proper-scoring sensitivity of the external validation). Because the survival-supervised information lives in W rather than H, scoring is a pure matrix projection requiring no retraining or validation survival information. Related supervised-NMF methods that instead route survival supervision through H require sample-loading re-estimation on each new cohort[32,33]. Validation survival outcomes were therefore used only for downstream performance evaluation, not for model training, scoring, or hyperparameter tuning. Reported external hazard ratios and log-rank statistics therefore reflect purely out-of-sample generalization. Further training and validation details appear in Supplementary Information, Part I.

To characterize the biological identity of the learned factors, we quantified how strongly each factor's gene loadings aligned with independently defined PDAC tumor and microenvironmental gene programs. For each factor, we correlated its gene loadings with binary membership indicators for each reference program (Spearman correlation over the top 50 genes per factor), assessing significance with Benjamini–Hochberg control of the false discovery rate within each factor. The same analysis was applied to DeSurv and to standard NMF fit at the same rank, and reference programs whose Spearman correlation exceeded 0.2 in at least one factor are shown in the main figure (full details in Supplementary Information). The matched-rank comparator shown in the figure is Lee multiplicative NMF (`NMF::nmf`), the field-standard implementation, which does not receive DeSurv's consensus initialization or hyperparameter tuning. We therefore also fit a matched-budget unsupervised control, which removes supervision and holds every other component of the DeSurv procedure fixed. Bayesian optimization was run with α pinned to 0 and the rank pinned to DeSurv's selected k, using the same optimizer, search budget, gene filter and cross-validation folds. The selected model was then fit by the same 100-restart consensus procedure. Any remaining difference in factor structure is therefore attributable to supervision rather than to tuning effort. A further sensitivity analysis instead held DeSurv's own λ, ν and n_{top} fixed and varied only α. Both controls are reported in Supplementary Information, Matched-budget unsupervised control at the same rank.

For orthogonal validation of D1 program identity, we evaluated the association between projected D1 scores and GATA6 RNA in situ hybridization measurements in the COMPASS cohort among the 33 tumors with both measurements, using Spearman correlation. Because *GATA6* is itself in the 810-gene projection support (805 of these genes were measured in the COMPASS expression data and contribute to the projected scores), part of this correlation could reflect the score being built from the assayed gene. We therefore repeated the comparison with *GATA6* removed from the projection, and the correlation was essentially unchanged (Spearman $\rho = 0.67$ versus 0.68), indicating the association is not an artifact of self-correlation.

Supervised comparator analyses

To distinguish prognostic discrimination from biological representation, we independently fit supervised principal components (superpc; cross-validated screening threshold and component number) and lasso-penalized Cox regression (glmnet; $\lambda_{\min}$) using the same 1,970-gene training universe as DeSurv. Each method performed feature selection and tuning using the training data only, after which the fitted model was frozen and applied to rank-transformed validation cohorts without refitting. The two comparators select features by different mechanisms, and neither matches DeSurv's. The penalized Cox comparator uses a pure lasso penalty (`glmnet` with $\alpha = 1$), whereas supervised principal components applies no penalty at all, instead screening genes by a cross-validated feature-score threshold and extracting a cross-validated number of components. DeSurv's internal Cox step uses an elastic net whose Bayesian-optimization-selected mixing parameter ($\xi = 0.056$) sits close to ridge. Both comparators are therefore far sparser in the genes they retain than DeSurv's 810-gene support, and both received a smaller tuning budget, a single internal cross-validation each against DeSurv's Bayesian-optimization search and consensus initialization, which should be borne in mind when reading their comparable external discrimination. No matched-budget control was run for these supervised comparators, as it was for the unsupervised arm; the comparison with them is therefore about what each representation resolves, not about tuned predictive performance. Correspondence between each supervised risk score and the DeSurv programs D1–D3 was quantified in the training cohort using absolute Spearman correlation. Additional analyses of selected-gene enrichment and compartment attribution are described in the Supplementary Information.

Simulation studies

Simulation studies were conducted to assess recovery of prognostic latent structure and survival prediction under controlled conditions. Gene expression data were generated from a nonnegative factor model $X = WH$, where gene loadings W comprised three gene classes: marker genes, background genes, and noise genes. Marker genes were simulated to load strongly on a single factor and weakly on others, background genes to load strongly across all factors, and noise genes to have uniformly low loadings. Specifically, marker gene primary loadings were drawn from $\text{Gamma}(\text{shape} = 3, \text{rate} = 0.8)$ and cross-loadings onto other factors from $\text{Gamma}(1, 20)$; background gene loadings from $\text{Gamma}(2, 1)$ across all factors; and noise gene loadings from $\text{Gamma}(1, 20)$. Sample-level factor activities H were drawn from $\text{Gamma}(2, 2)$. Each dataset comprised $G = 3{,}000$ genes, $N = 200$ samples, and $K = 3$ factors, with 150 marker genes per factor and 500 shared background genes. Cox regression coefficients were $\beta = (2, 0, 0)^\top$ in the prognostic scenario, $\beta = 0$ in the null scenario, and $\beta = (2, 0, 0)^\top$ in the mixed scenario where survival risk was driven by 300 genes comprising the 150 factor-1 markers and 150 randomly sampled background genes, creating partial overlap between the prognostic signal and variance-dominant outcome-neutral programs.

Survival times were generated from an exponential distribution in which risk depended on marker gene expression through $X^T W_{\text{true}}$, where W_{true} retained marker gene loadings for their corresponding factors and was zero otherwise; censoring times were generated independently from an exponential distribution. Each dataset was analyzed using DeSurv and standard NMF followed by Cox regression on inferred factors, both tuned using cross-validated concordance index via Bayesian optimization. Performance was summarized across repeated simulation replicates using test-set concordance and gene-program recovery. Gene-program recovery was quantified separately for each survival-driving factor. We first matched each true program to the learned factor whose top-ranked genes had the highest precision against it. Recovery was then the proportion of that learned factor's top-n_{top} genes belonging to the ground-truth prognostic gene set, with genes ranked by factor-specificity score and n_{top} set to the Bayesian-optimization-selected signature size. We report the average across survival-driving factors. In the primary scenario the target comprised the 150 factor-specific marker genes driving survival ($\beta = (2, 0, 0)$); in the mixed scenario, marker-only and full survival-gene definitions of the target were evaluated separately. We considered three simulation scenarios: a prognostic scenario in which prognostic programs explain low variance relative to outcome-neutral signals, a null scenario with no survival signal, and a mixed scenario where prognostic and variance-dominant programs partially overlap.

Real-world datasets

We analyzed publicly available and controlled-access RNA sequencing (RNA-seq) and microarray cohorts of PDAC with corresponding overall survival outcomes. Eligible samples were primary PDAC tumors with a measured bulk expression profile and valid overall-survival annotation, subject to cohort-specific quality-control and exclusion criteria detailed in Supplementary Information, Section 6; accrual and follow-up periods are as defined in each cohort's original publication. Gene expression matrices were analyzed on a log scale: RNA-seq cohorts as $\log_2(\text{TPM} + 1)$; microarray cohorts in platform-normalized log-scale intensities. For each training cohort separately, genes were ranked by mean expression and by variance independently; the top 3,000 genes with the lowest sum of these two ranks were retained, selecting genes that are both highly expressed and highly variable; the intersection across training cohorts yielded 1,970 genes used for all analyses. Validation datasets were restricted to this same gene set. Survival times and censoring indicators were taken from the associated clinical annotations. No missing-data imputation was performed: all analyses used complete cases, and analyses adjusted for the PurIST and DeCAF classifiers were restricted to samples with available classifier calls. Of the seven PDAC expression datasets we considered, two were used for training (TCGA and CPTAC) and five for external validation (Dijk, Moffitt, PACA-AU array, PACA-AU RNA-seq, and Puleo); because the two PACA-AU platforms profile one patient cohort, these five validation datasets span four independent patient cohorts. To harmonize differences in scale across cohorts, filtered gene expression data were within-sample rank transformed before model training (ranks are inherently nonnegative, preserving NMF compatibility); ranking removes platform- and library-size differences in expression scale that would otherwise dominate cross-cohort projection. More details about the datasets can be found in Supplementary Information, Section 6.

Per-cohort participant flow is summarized in Supplementary Table S1. Of 321 pooled training-cohort samples (TCGA-PAAD, $n = 181$; CPTAC-3, $n = 140$), 48 were excluded for missing survival time, missing event indicator, missing tumor grade, or quality-control failure, yielding 273 analytic training samples (TCGA-PAAD 144, CPTAC-3 129; 139 events). Of 979 pooled validation-cohort samples, 363 were excluded for non-PDAC histology, non-tumor tissue, or missing survival annotation, yielding 616 analytic validation profiles across five platform datasets: Dijk ($n = 90$, 81 events), Moffitt GEO array ($n = 123$, 83), PACA-AU array ($n = 63$, 38), PACA-AU RNA-seq ($n = 52$, 31), and Puleo array ($n = 288$, 181). PACA-AU included partially overlapping array and RNA-seq datasets (46 shared patients). Platform-specific analyses retained each dataset separately; for analyses combining validation datasets, duplicate patient accessions were represented once, preferentially retaining the RNA-seq profile and excluding the corresponding array profile, consistent with the preprocessing rule used in the previously published DeCAF analysis[5]. This yielded a combined validation population of 570 unique patients (388 deaths) spanning four independent patient cohorts.

Use of large language models

During preparation of this manuscript, the authors used a large language model (Claude; Anthropic) to assist with language editing and with code for generating and formatting the display items. All such output was directed, reviewed, and verified by the authors, who take full responsibility for the work. The model was not used to design the study, develop the DeSurv methodology, or interpret the findings, and it does not meet the criteria for authorship.

Data Availability

The training and validation datasets used in this study are publicly available or accessible through controlled-access repositories, and are de-identified; no Institutional Review Board (IRB) approval was required for their analysis. Training data were obtained from the Genomic Data Commons: TCGA-PAAD[43] and CPTAC-3[44]. Validation cohorts were obtained from ArrayExpress (Dijk, E-MTAB-6830[45]; Puleo, E-MTAB-6134[46]), the Gene Expression Omnibus (GEO; Moffitt, GSE71729[47]), and the International Cancer Genome Consortium (ICGC): the PACA-AU array cohort is publicly available from the Gene Expression Omnibus (GSE36924), and the PACA-AU RNA-seq cohort from the European Genome–phenome Archive (EGA) study EGAS00001000154 under controlled access[48]. Expression data and molecular classifications (PurIST, DeCAF) for the validation cohorts were originally compiled for the DeCAF study[5]; cohort details and sample sizes after quality control are summarized in Supplementary Information, Section 6. The Linehan FOLFIRINOX$_\pm$CCR2-inhibitor series, used for the paired pre- and post-treatment comparison (Fig. 4b), and the O'Kane/COMPASS treated metastatic PDAC cohort used both for the D1 prognostic-transportability survival analysis (Fig. 4a) and, in the subset of 33 tumors with matched staining, for the GATA6 RNA in situ hybridization comparison (main-text Fig. 2e), are restricted-access clinical-trial datasets that are not publicly deposited; raw data are available from the respective study investigators under a data-use agreement, subject to the governing trial consents and institutional approvals. To preserve reproducibility without redistributing restricted data, the derived per-cohort summary statistics reported in the main text and Supplementary Information are provided in the reproducibility repository (file `results/treated_cohort_stats.rds`), so that every treated-cohort value can be inspected and verified. Code used to generate the COMPASS D1–GATA6 comparison (main-text Fig. 2e) is also provided in the reproducibility repository; reproduction of the patient-level analysis requires authorized access to the COMPASS data.

Code Availability

All analyses were performed in R (version 4.6.0; R Core Team) using DeSurv (v1.0.1; this work), `NMF`[42], `glmnet`[23], `survival`, `ggplot2`, and `cowplot`; Bayesian optimization for hyper-parameter selection was implemented within DeSurv. The DeSurv R package is available at https://github.com/rashidlab/DeSurv (archived at Zenodo, DOI `10.5281/zenodo.20150929`). Reproducibility code, processed data, and the exact session-info file are available at https://github.com/rashidlab/DeSurv-paper (archived at Zenodo, DOI `10.5281/zenodo.20150946`); this includes `code/11_treated_cohort_analysis.R`, which generates the treated-cohort summary statistics above.

Acknowledgements

This work was supported by the National Cancer Institute grants U01 CA274298, P50 CA257911, T32 CA106209, and R01 CA288145, and by the Department of Defense Peer Reviewed Cancer Research Program (HT9425241103100121). We thank the patients and research teams who contributed samples and clinical data to the public cohorts used here (TCGA, CPTAC, Dijk, Moffitt, Puleo, PACA-AU).

Inclusion and Diversity. The patient cohorts analyzed in this study were curated by the original investigators or consortia (TCGA, CPTAC, ICGC, and the original Moffitt, Dijk, and Puleo studies) and our access was limited to the de-identified expression and survival data they provide. Sex was recorded in every cohort and was included as a covariate in the clinical-adjustment analyses (Supplementary Table S4), where the prognostic association of the DeSurv linear predictor was unchanged. We did not conduct sex-stratified analyses, because the study was not powered for subgroup comparisons within individual cohorts, and we did not stratify by race or ethnicity because those annotations were not consistently available across cohorts. Whether the prognostic relevance of these gene programs differs by sex, race, or ethnicity remains an important question for adequately powered, more comprehensively annotated cohorts.

Author Contributions

Conceptualization: A.M.Y., N.U.R. Methodology: A.M.Y., A.Y., D.L., N.U.R. Software: A.M.Y. Formal analysis: A.M.Y. Investigation: A.M.Y. Data curation: A.M.Y., X.L.P. Visualization: A.M.Y. Writing – original draft: A.M.Y., N.U.R. Writing – review and editing: A.M.Y., A.Y., X.L.P., J.J.Y., D.L., N.U.R. Resources (PDAC domain expertise and biological interpretation): X.L.P., J.J.Y. Supervision: N.U.R. Funding acquisition: A.M.Y., N.U.R. All authors reviewed and approved the final manuscript.

Competing Interests

N.U.R. and J.J.Y. are named as inventors on US patents US20220127676A1 and US20250066856A1, which relate to pancreatic cancer subtype classification. The PurIST and DeCAF classifiers were used only as published, independently derived comparators: their subtype calls were obtained with the released classifiers and included as fixed covariate annotations, and the adjustment analyses using these calls are objective and reproducible from the deposited code. They are not components of DeSurv. The other authors declare no competing interests.

References

1. Siegel, R. L., Kratzer, T. B., Wagle, N. S., Sung, H. & Jemal, A. Cancer statistics, 2026. *CA: A Cancer Journal for Clinicians* https://doi.org/10.3322/caac.70043 (2026) doi:10.3322/caac.70043.
2. Collisson, E. A. *et al.* Subtypes of pancreatic ductal adenocarcinoma and their differing responses to therapy. *Nature Medicine* **17**, 500–503 (2011).
3. Moffitt, R. A. *et al.* Virtual microdissection identifies distinct tumor- and stroma-specific subtypes of pancreatic ductal adenocarcinoma. *Nature Genetics* **47**, 1168–1178 (2015).
4. Rashid, N. U. *et al.* Purity independent subtyping of tumors (PurIST), a clinically robust, single-sample classifier for tumor subtyping in pancreatic cancer. *Clinical Cancer Research* **26**, 82–92 (2020).
5. Peng, X. L. *et al.* DeCAF defines clinical fibroblast subtypes and multidimensional tumor-stroma crosstalk shaping prognosis and immunotherapy response. *Cell Reports Medicine* **7**, 102611 (2026).
6. Moncada, R. *et al.* Integrating microarray-based spatial transcriptomics and single-cell RNA-seq reveals tissue architecture in pancreatic ductal adenocarcinomas. *Nature Biotechnology* **38**, 333–342 (2020).
7. Hwang, W. L. *et al.* Single-nucleus and spatial transcriptome profiling of pancreatic cancer identifies multicellular dynamics associated with neoadjuvant treatment. *Nature Genetics* **54**, 1178–1191 (2022).
8. Werba, G. *et al.* Single-cell RNA sequencing reveals the effects of chemotherapy on human pancreatic adenocarcinoma and its tumor microenvironment. *Nature Communications* **14**, 797 (2023).
9. Bair, E. & Tibshirani, R. Semi-supervised methods to predict patient survival from gene expression data. *PLoS Biology* **2**, e108 (2004).
10. Aung, K. L. *et al.* Genomics-driven precision medicine for advanced pancreatic cancer: Early results from the COMPASS trial. *Clinical Cancer Research* **24**, 1344–1354 (2018).
11. Nguyen, H., Nguyen, H., Tran, D., Draghici, S. & Nguyen, T. Fourteen years of cellular deconvolution: Methodology, applications, technical evaluation and outstanding challenges. *Nucleic Acids Research* **52**, 4761–4783 (2024).
12. Lee, D. D. & Seung, H. S. Learning the parts of objects by non-negative matrix factorization. *Nature* **401**, 788–791 (1999).
13. Brunet, J.-P., Tamayo, P., Golub, T. R. & Mesirov, J. P. Metagenes and molecular pattern discovery using matrix factorization. *Proceedings of the National Academy of Sciences USA* **101**, 4164–4169 (2004).
14. Peng, X. L., Moffitt, R. A., Torphy, R. J., Volmar, K. E. & Yeh, J. J. De novo compartment deconvolution and weight estimation of tumor samples using DECODER. *Nature Communications* **10**, 4729 (2019).
15. Bailey, P. *et al.* Genomic analyses identify molecular subtypes of pancreatic cancer. *Nature* **531**, 47–52 (2016).
16. Cook, R. D. Fisher lecture: Dimension reduction in regression. *Statistical Science* **22**, 1–26 (2007).

17. Torre-Healy, L. A. *et al.* Open-source curation of a pancreatic ductal adenocarcinoma gene expression analysis platform (pdacR) supports a two-subtype model of clinical relevance. *Communications Biology* **6**, 163 (2023).

18. Aran, D., Sirota, M. & Butte, A. J. Systematic pan-cancer analysis of tumour purity. *Nature Communications* **6**, 9971 (2015).

19. Donoho, D. & Stodden, V. When does non-negative matrix factorization give a correct decomposition into parts? in *Advances in neural information processing systems 16 (NIPS 2003)* (eds. Thrun, S., Saul, L. K. & Schölkopf, B.) 1141–1148 (MIT Press, 2004).

20. Laurberg, H., Christensen, M. G., Plumbley, M. D., Hansen, L. K. & Jensen, S. H. Theorems on positive data: On the uniqueness of NMF. *Computational Intelligence and Neuroscience* **2008**, 764206 (2008).

21. Gillis, N. The why and how of nonnegative matrix factorization. in *Regularization, optimization, kernels, and support vector machines* (eds. Suykens, J. A. K., Signoretto, M. & Argyriou, A.) 257–291 (Chapman; Hall/CRC, 2014).

22. Lautizi, M., Baumbach, J., Weichert, W., Steiger, K. & List, M. A systematic assessment of pancreatic cancer molecular subtype classification across cohorts. *NAR Cancer* **4**, zcac030 (2022).

23. Simon, N., Friedman, J. H., Hastie, T. & Tibshirani, R. Regularization paths for cox's proportional hazards model via coordinate descent. *Journal of Statistical Software* **39**, 1–13 (2011).

24. Raphael, B. J. *et al.* Integrated genomic characterization of pancreatic ductal adenocarcinoma. *Cancer Cell* **32**, 185–203 (2017).

25. Cao, L. *et al.* Proteogenomic characterization of pancreatic ductal adenocarcinoma. *Cell* **184**, 5031–5052 (2021).

26. Snoek, J., Larochelle, H. & Adams, R. P. Practical Bayesian optimization of machine learning algorithms. in *Advances in neural information processing systems 25 (NeurIPS 2012)* (2012).

27. Shahriari, B., Swersky, K., Wang, Z., Adams, R. P. & Freitas, N. de. Taking the human out of the loop: A review of Bayesian optimization. *Proceedings of the IEEE* **104**, 148–175 (2016).

28. Frigyesi, A. & Höglund, M. Non-negative matrix factorization for the analysis of complex gene expression data: Identification of clinically relevant tumor subtypes. *Cancer Informatics* **6**, 275–292 (2008).

29. Leary, J. R. *et al.* Sub-cluster identification through semi-supervised optimization of rare-cell silhouettes (SCISSORS) in single-cell RNA-sequencing. *Bioinformatics* **39**, btad449 (2023).

30. Maurer, C. *et al.* Experimental microdissection enables functional harmonisation of pancreatic cancer subtypes. *Gut* **68**, 1034–1043 (2019).

31. Shapley, L. S. A value for n-person games. in *Contributions to the theory of games, volume II* (eds. Kuhn, H. W. & Tucker, A. W.) 307–317 (Princeton University Press, 1953).

32. Huang, Z., Salama, P., Shao, W., Zhang, J. & Huang, K. Low-rank reorganization via proportional hazards non-negative matrix factorization unveils survival associated gene clusters. *Preprint at https://arxiv.org/abs/2008.03776* (2020).

33. Le Goff, V. *et al.* SurvNMF: Non-negative matrix factorization supervised for survival data analysis. *Preprint at https://hal.science/hal-04975434* (2025).

34. Zaccaria, G. M., Altini, N., Mezzolla, G., *et al.* SurvIAE: Survival prediction with interpretable autoencoders from diffuse large B-cell lymphoma gene expression data. *Computer Methods and Programs in Biomedicine* **244**, 107966 (2024).

35. Meng, X., Wang, X., Zhang, X., *et al.* A novel attention-mechanism based Cox survival model by exploiting pan-cancer empirical genomic information. *Cells* **11**, 1421 (2022).

36. Ogawa, Y., Masugi, Y., Abe, T., *et al.* Three distinct stroma types in human pancreatic cancer identified by image analysis of fibroblast subpopulations and collagen. *Clinical Cancer Research* **27**, 107–119 (2021).

37. Neuzillet, C., Nicolle, R., Raffenne, J., *et al.* Periostin- and podoplanin-positive cancer-associated fibroblast subtypes cooperate to shape the inflamed tumor microenvironment in aggressive pancreatic adenocarcinoma. *The Journal of Pathology* **258**, 408–425 (2022).

38. Lee, J. J., Kearney, J. F., Trembath, H. E., *et al.* Tumor-intrinsic and cancer-associated fibroblast subtypes independently predict outcomes in pancreatic cancer. *Annals of Surgery* **280**, 659–666 (2024).

39. Newman, A. M., Steen, C. B., Liu, C. L., Gentles, A. J., *et al.* Determining cell type abundance and expression from bulk tissues with digital cytometry. *Nature Biotechnology* **37**, 773–782 (2019).

40. Chu, T., Wang, Z., Pe'er, D. & Danko, C. G. Cell type and gene expression deconvolution with BayesPrism enables Bayesian integrative analysis across bulk and single-cell RNA sequencing in oncology. *Nature Cancer* **3**, 505–517 (2022).

41. Smith, A. M., Walsh, J. R., Long, J., *et al.* Standard machine learning approaches outperform deep representation learning on phenotype prediction from transcriptomics data. *BMC Bioinformatics* **21**, 119 (2020).

42. Gaujoux, R. & Seoighe, C. A flexible r package for nonnegative matrix factorization. *BMC Bioinformatics* **11**, 367 (2010).

43. The Cancer Genome Atlas Research Network. Data from 'integrated genomic characterization of pancreatic ductal adenocarcinoma' (TCGA-PAAD). (2017).

44. National Cancer Institute Clinical Proteomic Tumor Analysis Consortium. Data from CPTAC-3 pancreatic ductal adenocarcinoma cohort. (2019).

45. Dijk, F. *et al.* Data from 'unsupervised class discovery in pancreatic ductal adenocarcinoma'. (2020).

46. Puleo, F. *et al.* Data from 'stratification of pancreatic ductal adenocarcinomas based on tumor and microenvironment features'. (2018).

47. Moffitt, R. A. *et al.* Data from 'virtual microdissection identifies distinct tumor- and stroma-specific subtypes of pancreatic ductal adenocarcinoma'. (2015).

48. Australian Pancreatic Cancer Genome Initiative & International Cancer Genome Consortium. Pancreatic cancer (PACA-AU) cohort: Array and RNA-sequencing data. (2016).

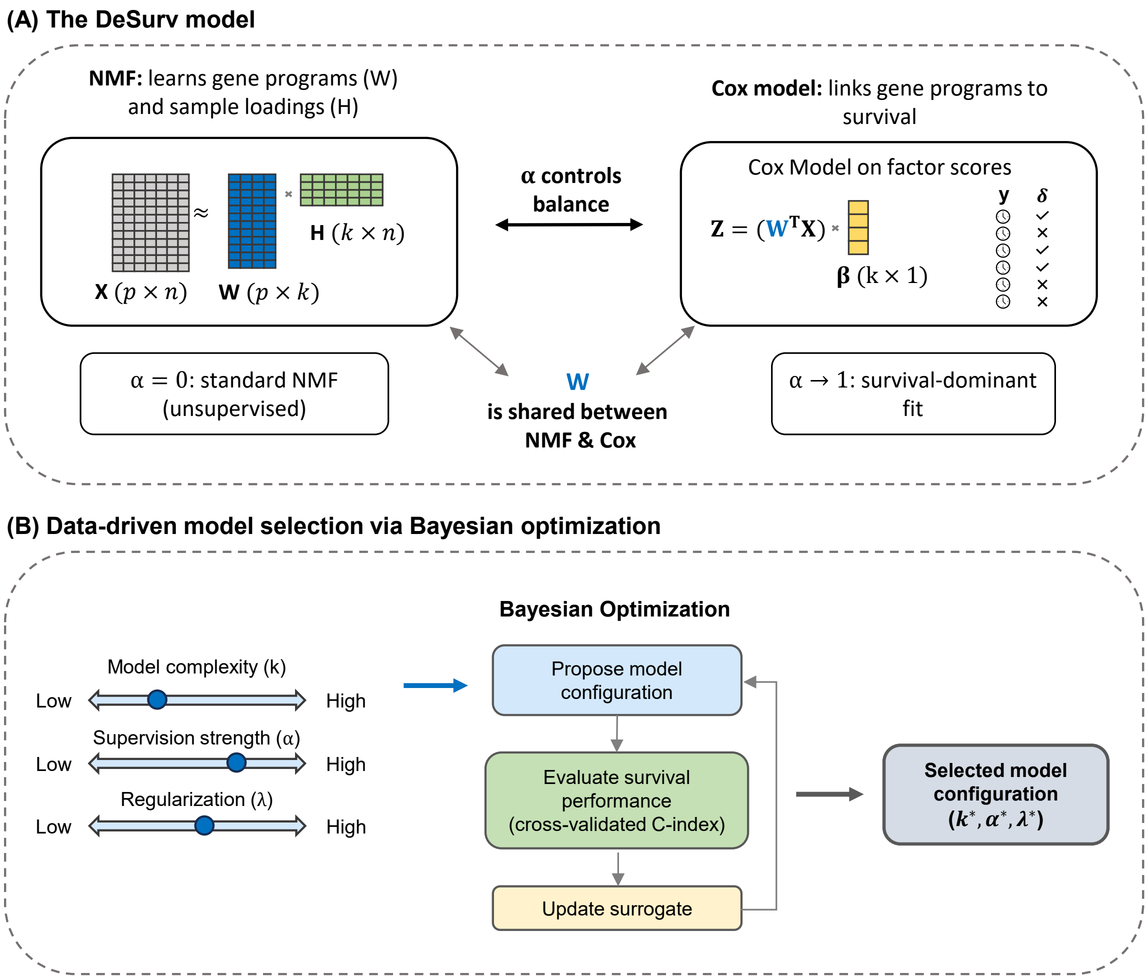

Figure 1: Overview of the DeSurv framework. (A) DeSurv jointly optimizes NMF reconstruction ($X \approx WH$) and Cox proportional hazards likelihood. The gene program matrix W is shared between objectives: factor scores $Z = W^\top X$ serve as covariates in the Cox model with coefficients β. A supervision parameter $\alpha \in [0, 1)$ controls the balance between reconstruction fidelity ($\alpha = 0$, standard NMF) and survival prediction (α close to 1). (B) The factorization rank (k), supervision strength (α), and regularization (λ) are selected via Bayesian optimization over cross-validated concordance index. Survival supervision acts on the gene-program matrix W during discovery, rather than being applied only after an unsupervised factorization has been completed; following training-only model selection, the resulting gene weights are fixed for direct evaluation in external cohorts.

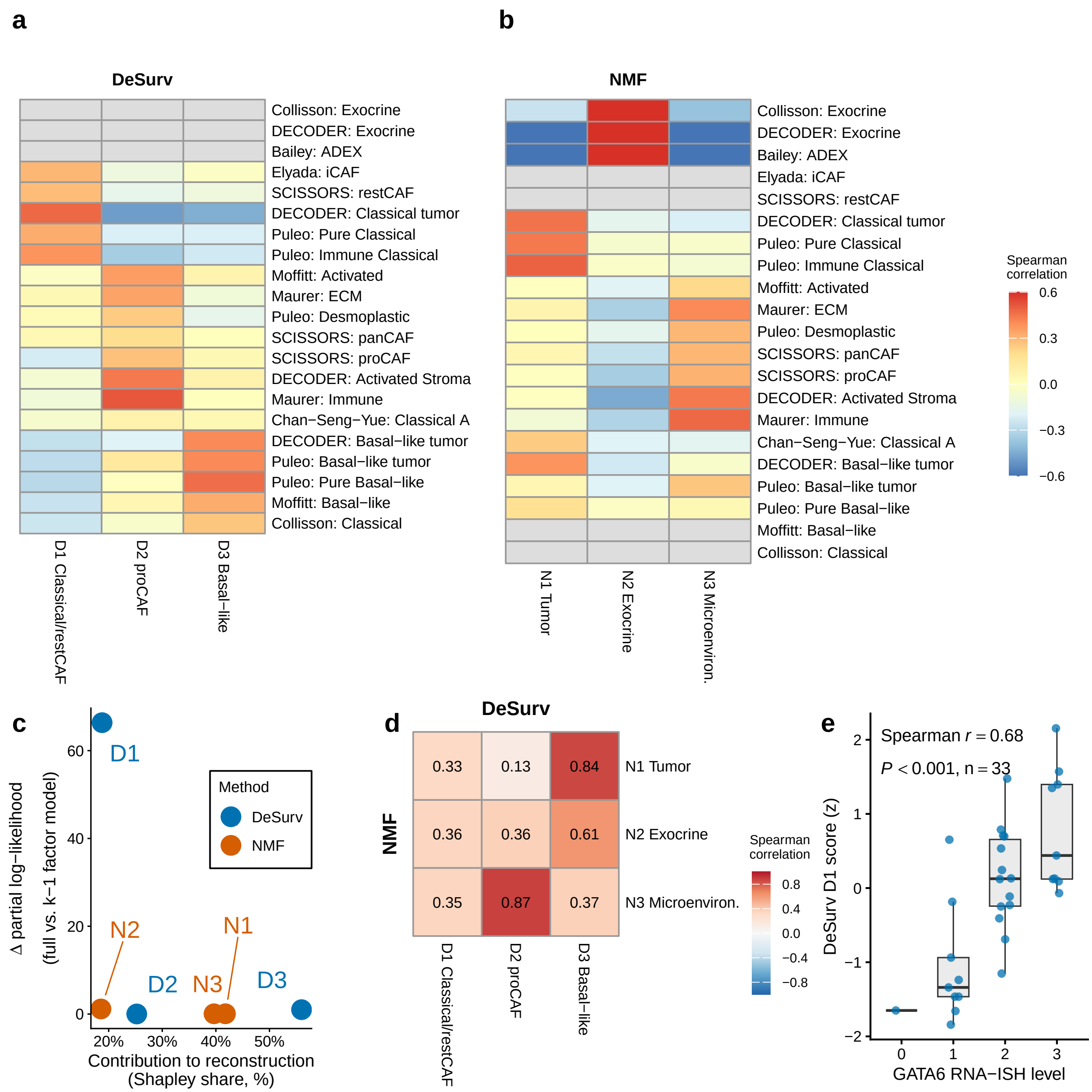

Figure 2: Survival supervision produces different factor structures in PDAC (TCGA + CPTAC training cohort, $n = 273$ patients, 139 events). (a,b) Spearman correlations between factor gene rankings (top 50 genes per factor) and established PDAC gene programs for DeSurv (a) and standard NMF (b) at $k = 3$. Both panels display the same rows, the union of reference programs correlating at $r > 0.2$ with at least one factor of either method; grey cells denote programs with none of their genes in that method's displayed gene space, where the correlation is not computable. The rows labelled SCISSORS: restCAF and SCISSORS: proCAF display the SCISSORS iCAF and myCAF gene sets under the naming used by the DeCAF classification (Supplementary Information). (c) Per-factor reconstruction contribution (normalized Shapley share, summing to 100%) versus survival contribution (Δ Cox partial log-likelihood on factor removal). (d) Pairwise correspondence between NMF and DeSurv factors (Spearman correlation of W-matrix gene loadings over all genes). (e) DeSurv D1 score versus GATA6 RNA in situ hybridization level in the COMPASS cohort (Spearman correlation; $n = 33$).

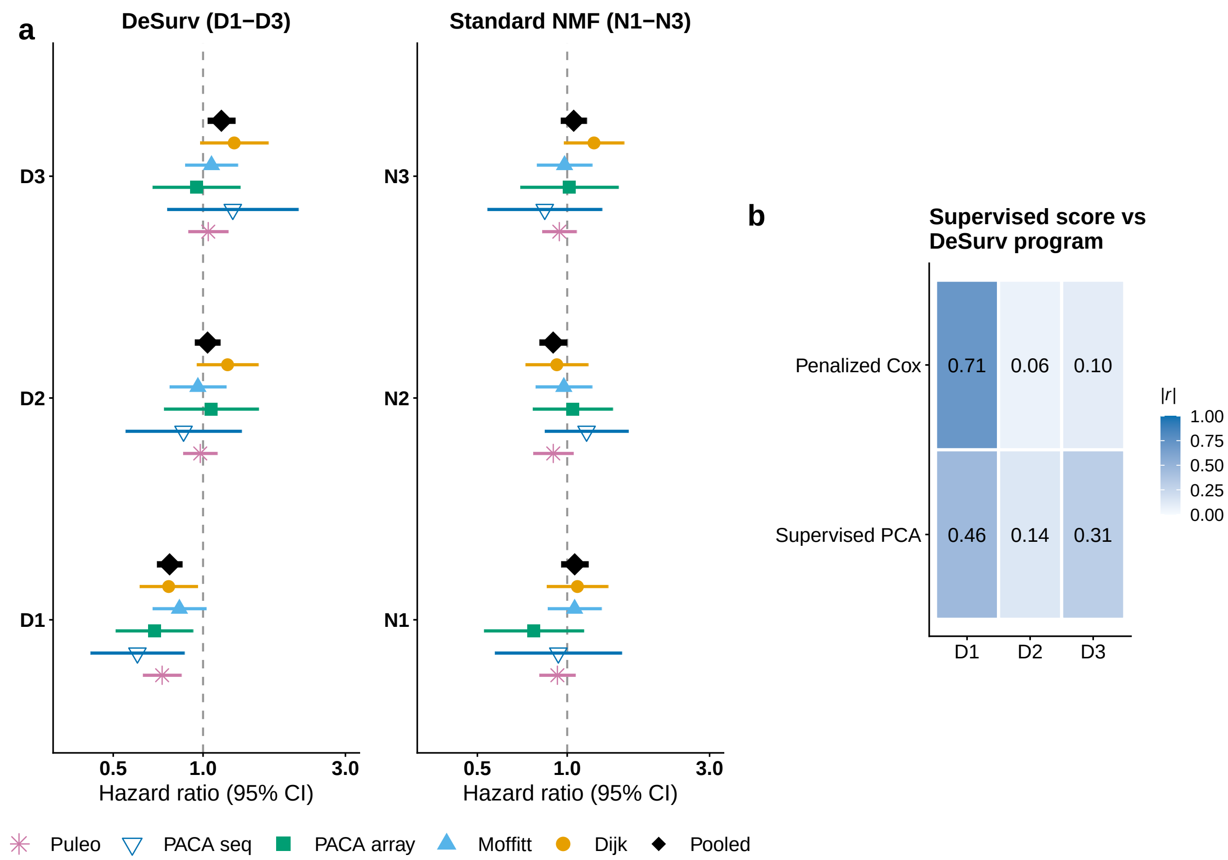

Figure 3: Survival-aligned programs generalize across independent PDAC patient cohorts (five external validation datasets: Dijk, Moffitt GEO array, PACA-AU array, PACA-AU RNA-seq and Puleo; combined $n = 570$ unique patients, 388 deaths, with PACA-AU patients profiled on both platforms represented once via RNA-seq). (a) Forest plot of per-dataset univariate hazard ratios (95% CIs) for each factor; the platform-specific PACA-AU array and RNA-seq estimates are shown separately, and the black diamonds denote the combined estimate over unique patients (per-factor cohort-stratified Cox on the de-duplicated patient-level data; PACA-AU array duplicates excluded). Puleo is the largest contributor, supplying 288 of the 570 unique patients (51%), which is why the combined estimate is also reported with each dataset omitted in turn; those leave-one-dataset-out estimates range from 1.34 to 1.45. (b) Correspondence (|Spearman r|) of tuned supervised risk scores (penalized Cox and supervised PCA) with the three DeSurv programs (D1–D3); supervised scores concentrate on D1 and are only weakly correlated with the D2 stromal program. No validation-cohort survival outcomes were used in training, scoring or hyperparameter selection.

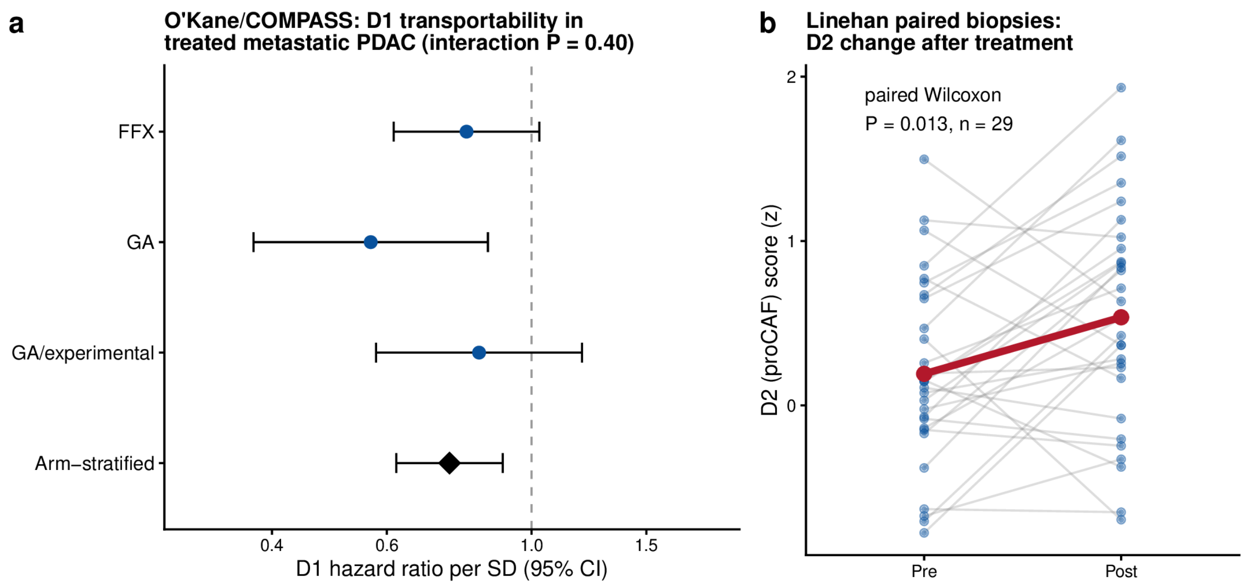

Figure 4: Transportability of the frozen DeSurv programs into independently treated PDAC cohorts. (a) Prognostic transportability of D1 in treated metastatic PDAC. Forest plot of hazard ratios per SD of the frozen D1 score within each O'Kane/COMPASS treatment arm (FOLFIRINOX, gemcitabine/nab-paclitaxel [GA], and GA/experimental); the diamond is the common association from an arm-stratified Cox model with treatment-specific baseline hazards (not a pooled-arm estimate), HR 0.75 (95% CI 0.62–0.90, $P = 0.003$). D1 was protective in all three arms; the arm-by-score interaction was not significant (interaction $P = 0.40$), though the individual arms are underpowered for detecting heterogeneity. (b) The proCAF program D2 before and after treatment in paired Linehan biopsies ($n = 29$, paired Wilcoxon $P = 0.013$); grey lines are individual patients, the red line the group mean (+0.34 SD). Error bars denote 95% confidence intervals. Because treatment, stage and cohort were not independently varied, these analyses establish prognostic and biological transportability rather than treatment-predictive effects.

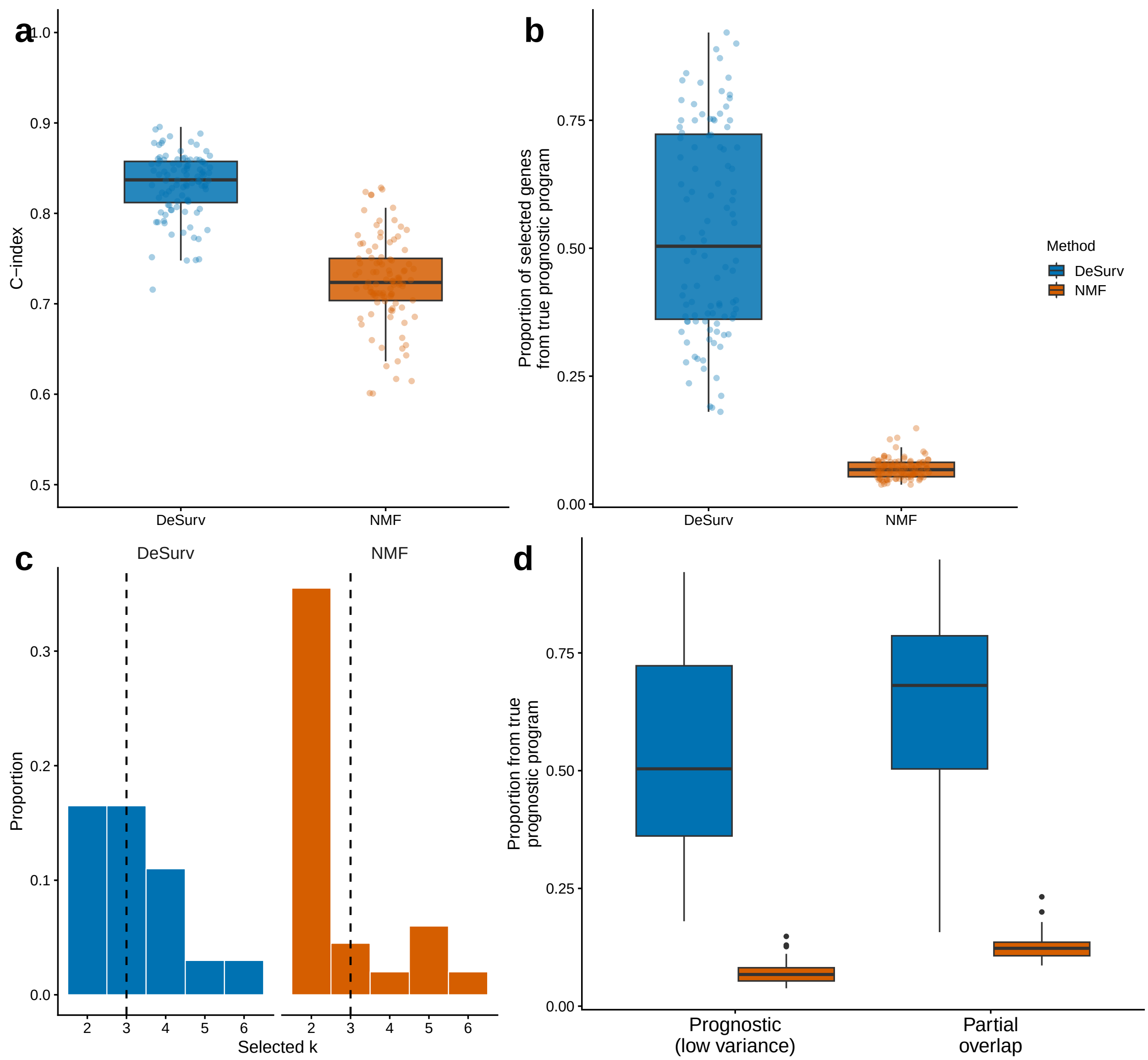

Figure 5: Survival supervision improves recovery of prognostic gene programs when variance and prognosis diverge. (a) Test-set concordance across 100 simulation replicates comparing DeSurv with standard NMF ($\alpha = 0$; $p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$; paired Wilcoxon $P < 0.001$). (b) Recovery of genes defining the true prognostic program, quantified as the proportion of selected genes in the best-matching learned program that belonged to the ground-truth prognostic gene set. In (a) and (b), boxes show median and interquartile range; whiskers extend to 1.5×IQR; points are individual replicates. (c) Distribution of selected ranks across replicates for DeSurv versus standard NMF. (d) The same gene-program recovery metric as in (b), computed for two signal regimes: a prognostic program that explains low transcriptomic variance (separated from the dominant variance) versus one that partially overlaps the dominant variance. DeSurv recovers the prognostic program in both regimes, whereas standard NMF, which follows variance, does not; the null scenario (no prognostic program) is shown in the Supplementary Information. These scenarios are constructed to isolate the regime in which the prognostic program carries low transcriptomic variance, and therefore quantify recovery under that construction; the concordance gap in (a) is not an expected discrimination gain in real data, where the two approaches perform comparably (Table 1).